

		(ii) As volume compresses, the <u>time</u> between successive collisions of a molecule <u>against the wall decreases</u> or the <u>frequency</u> of the collisions of molecules <u>against the wall increases</u>. For same change in linear momentum, the <u>net force acting on the wall increases</u>, hence increasing pressure	B1 B1
	(b)	(i) $pV = nRT$ For A, $1.5 \times 10^5 \times 2.0 \times 10^{-3} = n \times 8.31 \times 200$ $n = 0.18 \text{ mol}$	A1
		(ii) $E = \frac{1}{2} m c_{rms}^2 = \frac{3}{2} kT$ $\frac{1}{2} \times 40 \times 1.66 \times 10^{-27} \times c_{rms}^2 = \frac{3}{2} \times 1.38 \times 10^{-23} \times 200$ $c_{rms} = 353 \approx 350 \text{ m s}^{-1}$	M1 A1
		(iii) For C to D $\Delta U = Q + W$ $\frac{3}{2} (0.181)(8.31)(400 - 720) = -50 + W$ $W = -670 \text{ J}$ Work is done by the gas	M1 A1 A1
3	(a)	The periodic motion of an object about the equilibrium point in which its acceleration	B1